Bi-NAC Paper Review: 6-Minute Script

Target pace: about 155 words per minute (a brisk but clear pace). Total: about 930 words, or about 6:00. **Bold** marks words to stress. Running time is in brackets.

Slide 1 — Title [0:00–0:15]

Hi everyone. Today I'm reviewing **"RL with Learnable Textual Feedback: A Bilevel Approach,"** by Singh, Sredharan, Chakraborty and Bedi, from UCF, CMU and the University of Maryland. It was accepted at **ICML 2026**.

Slide 2 — Contents [0:15–0:22]

I'll cover the problem, the paper's novelty and approach, the algorithm, where it sits relative to the state of the art, and finally the evaluation.

Slide 3 — The problem: sparse rewards [0:22–1:05]

RL with verifiable rewards, like **GRPO**, behind DeepSeek-R1, is now the standard way to train LLMs to reason. But the reward is **binary**: one if the final answer is right, zero otherwise. A mostly-right answer gets nothing.

GRPO compares answers within a group, so if **every** answer is wrong, every advantage is zero and **nothing is learned**. That's **advantage collapse**.

The authors measured it on MBPP with LLaMA-3.2-1B. **74%** of training groups collapsed, and **83%** of answers got zero reward. In the plot you can see vanilla GRPO stuck in a **zero-reward phase for about 600 steps**.

Slide 4 — Why this problem matters [1:05–1:40]

So, quoting the paper, RL for reasoning **needs denser signals than terminal correctness**. The common fix is a **critic LLM** that writes feedback in natural language.

The gap: existing critics are **frozen** or trained on objectives **unrelated to the actor**. On MATH-500, RL lifts accuracy from 26.2 to 36.6%, and a fixed critic only reaches **39.6%**. As the highlighted text says, a critique can be **fluent and even correct, yet still fail to improve the actor**. Bi-NAC reaches **46.6%**.

Slide 5 — Novelty and contributions [1:40–2:20]

The key insight: **judge feedback by its effect on the actor**. The question changes from "*is this critique correct?*" to "*does this critique raise the actor's reward once the actor has learned to use it?*"

These two problems depend on each other: a critique's usefulness depends on **how the actor adapts**, and the actor's adaptation depends on **what feedback it gets**. **The critic and actor must be trained together.** The paper calls this a new paradigm: **learnable textual feedback for RL**.

Slide 6 — Bi-NAC approach [2:20–3:00]

Bi-NAC, the Bilevel Natural Language Actor-Critic, models this as a **Stackelberg game**: a leader moves first and a follower responds.

Here the **critic is the leader**. It writes feedback, anticipating how the actor will learn from it. The **actor is the follower**. It learns to use the feedback to earn the verified reward.

Two things make this different from a classic actor-critic. First, the critic **generates text**; it doesn't predict values. Second, it is **trained through the actor's learning dynamics**. Its reward is the actor's reward *after* the actor revises. In the figure, the unaligned critic on the left leaves the answer wrong; the aligned one on the right fixes it.

Slide 7 — Algorithm and math model [3:00–3:50]

Each training iteration starts with a **two-turn rollout**. The actor makes a first attempt, y-zero. The critic writes feedback, z. The actor revises to y-one, and a verifier scores it zero or one.

Formally, the critic maximizes reward subject to the actor being optimal for it, and the authors turn this nested problem into a single **Lagrangian penalty** with multiplier lambda.

That gives two simple updates. The **actor** gets a standard **policy gradient on both turns**, so it learns to make good first attempts *and* to use the feedback. The **critic** is rewarded only when its feedback **leads to a correct revision**. Its gradient has two terms. **G-one** reinforces feedback that helps the *current* actor. **G-two** subtracts the same quantity computed with a **reference actor**, a recent checkpoint standing in for the optimal actor. That subtraction is the bilevel penalty that ties the critic to how the actor is learning.

Slide 8 — Position vs. state of the art [3:50–4:30]

Bi-NAC draws on four lines of research. From **RL with verifiable rewards** it takes the simple, scalable reward, but that line suffers advantage collapse. From **critique-guided RL**, like Critique-GRPO, it takes the LLM critic, but those critics are frozen or trained apart from the actor. From **self-correction** methods, like SCoRe and Reflexion, it takes the attempt-then-revise loop, but there the feedback is generic and only the actor learns. And from **bilevel optimization and meta-learning** it takes the idea of training one learner through another's learning, which so far had only been used for **scalar** rewards.

Bi-NAC is the **first to train a critic that writes text jointly with the actor**, rewarding the critic by the actor's improvement.

Slide 9 — Does Bi-NAC beat SOTA? [4:30–5:10]

They tested on MATH-500, MBPP, GPQA and AIME, with LLaMA and Qwen models from 1 to 8 billion parameters. Panel A: Bi-NAC beats the strongest baseline at **every model size on every benchmark**, by 2.5 to **11.7 points**, with GPQA at 8B going from 44.6 to **56.3%**. Panel B: fixed or auxiliary critics all **plateau around 38 to 40%**, while Bi-NAC reaches **46.6**. Panel C: **smaller beats bigger**. A 2B Bi-NAC beats a 3B GRPO model, and 6B beats 7B. Panel D: on very hard competition math, one feedback round **doubles** AIME 2024 accuracy.

Slide 10 — Evaluation and validation [5:10–5:55]

Does the evidence support the claim? The ablation in panel E is the key result. Training only one model gives 38 to 40%. Training both **without** the bilevel term gives 41.8%. **Adding the coupling gives 46.6%**, and the LLM-judged feedback actionability doubles, from 2.1 to **4.2**. So the coupling itself is what drives the gain.

Gains keep growing **up to three revision rounds** and then hold steady. Training is cheaper too: **3.2 times fewer tokens** and 11% less training time than GRPO 3B. The cost is about **17% slower inference**. It works across mixed LLaMA and Qwen pairs, and **one model can play both roles**.

My verdict: the result is **validated in direction**. But there are no error bars, AIME has only 30 problems, and feedback quality is scored by an LLM, so the exact size of the gains deserves some caution.

Slide 11 — Thank you [5:55–6:00]

Thank you. I'm happy to take questions.

Likely questions, with quick answers

- **Why is it "bilevel" if the critic and actor maximize the same reward?** Both objectives are the actor's final reward, so the game is cooperative. The bilevel structure comes from the critic's constraint that the actor is optimal for it. In practice that becomes the penalty G-two, computed against a checkpoint actor rather than a true optimal actor.
- **Isn't "2B vs. 3B" unfair, since Bi-NAC runs two models?** The single-model 1B variant scores 46.8%, so the gain holds even at 1B parameters.
- **What is FO?** Feedback actionability: an LLM judge scores each critique from 1 to 5 for precision, relevance and actionability, grading all methods together.
- **Why train with only two turns?** Most of the gain comes at turn 2 (+7.8), and accuracy stays flat after turn 3, so two turns give the best gain for the compute.